Assessment Scenario

Choose **one of the following AI application domains**

- Healthcare diagnostics and predictive analytics
- Financial credit scoring or fraud detection
- Recruitment and HR decision systems
- Smart city surveillance and traffic management
- Retail personalization and advertising algorithms </aside>

You will evaluate the **ethical, legal, and security challenges** inherent in the chosen application and propose a comprehensive framework for responsible AI deployment.

Assessment Tasks

Task 1: Ethical and Fairness Evaluation

- Identify potential sources of bias within the AI system's data, model, or deployment process.
- Apply relevant **fairness metrics** (e.g., demographic parity, equalized odds) to discuss how bias could be measured or mitigated.
- Explain how principles of **Fairness, Accountability, Transparency, and Privacy (FAT/ML)** are applied in your chosen use case.
- Reference one real-world case study (e.g., COMPAS, Amazon, or Twitter/X) to compare your analysis.

Task 2: Governance and Accountability Framework

- Propose an **AI governance model** for your organization, including roles, reporting lines, and oversight mechanisms.
- Outline policies for model documentation, auditing, and stakeholder communication (e.g., **Model Cards**, **Datasheets for Datasets**).
- Discuss how the system aligns with **global regulatory frameworks** such as GDPR, CCPA, and the EU AI Act.
- Highlight how corporate responsibility and ethics committees contribute to long-term accountability. </aside>

Task 3: Data Security and Privacy Strategy

- Design a **data security architecture** outlining safeguards across the AI lifecycle (collection, storage, training, deployment).
- Explain how encryption, anonymization, or federated learning can protect user data.
- Identify the **monitoring tools and metrics** you would use to assess ongoing security performance (e.g., SIEM, IDS, or drift detection).
- Address compliance with privacy-by-design principles and cross-border data protection requirements.

Task 4: Risk and Continuity Plan

- Conduct a **threat modeling analysis** to identify critical risks to AI integrity (data poisoning, model theft, adversarial attacks).
- Describe how business continuity would be maintained if AI systems were compromised or went offline.
- Propose AI-driven strategies for early detection, incident response, and recovery.
- Reflect on the balance between **innovation and control**, referencing lessons from Tesla's cybersecurity challenges.

Deliverables

1. **Executive Summary** – Concise overview of the system, risks, and recommendations.
2. **Main Body** – Structured analysis covering the four assessment tasks above.
3. **Conclusion** – Reflections on how ethical design, security, and governance interconnect in sustainable AI practices.
4. **References** – All cited works in APA or Harvard format.

Optional: Include diagrams or tables (e.g., AI lifecycle maps, governance frameworks, or threat matrices) to strengthen clarity.

Grading

Marks for each of the criteria will be assigned accordingly:

- 0 - Failed to submit
- 1 - Limited
- 2 - Basic
- 3 - Proficient
- 4 - Advanced
- 5 - Exceptional

The following assessment criteria will be used to grade your submission:

- Effort
- Application
- Examples
- AI for Business insight

Rubric

Criteria	1	2	3	4	5
	Limited	**Basic**	**Proficient**	**Advanced**	**Exceptional**
Effort	Minimal effort apparent, with little attention to detail or engagement with the task.	Shows some effort, but responses may lack polish or thoroughness.	Demonstrates a solid effort, with responses showing attention to detail and engagement with the task.	Displays a strong effort, with responses exhibiting thoroughness, coherence, and depth.	Exemplifies an outstanding effort, with responses displaying exceptional clarity, insight, and completeness.
Application	Fails to apply learned information effectively to the scenario.	Demonstrates basic application of learned information to the scenario, but lacks depth or coherence.	Applies learned information to the scenario with clarity and coherence, demonstrating a solid understanding of its relevance.	Applies learned information skillfully to the scenario, offering insightful analysis and connections.	Applies learned information in a highly sophisticated manner, integrating multiple perspectives and drawing nuanced conclusions.
Examples	Few or no examples provided. Demonstrates limited understanding or application of concepts.	Provides some examples, but they may be simplistic or lack depth. Demonstrates basic understanding of concepts.	Offers several relevant examples demonstrating understanding of concepts. Examples are well-integrated into the response.	Presents numerous detailed examples that vividly illustrate concepts and their applications.	Presents numerous detailed examples that vividly illustrate concepts and their applications.
AI for Business Insight	Shows little understanding of the principles of AI for Business Strategy and fails to connect theories to the scenario.	Displays basic understanding of AI for Business Strategy; makes simple, expected connections to the scenario.	Demonstrates a clear understanding of AI for Business Strategy and makes relevant connections to the scenario, reflecting solid insight.	Exhibits deep understanding of AI for Business Strategy concepts; offers thoughtful, original insights that enhance the scenario analysis.	Demonstrates an understanding of AI for Business while; integrating sophisticated insights that significantly enrich the scenario, reflecting a high level of critical thinking.